SocietyBench: Forecasting Counterfactual Social-World Evolution

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Project page co-minder.github.io/Societybench

Abstract

Large language models (LLMs), and the agents built on top of them, are now benchmarked heavily on whether they can *finish a task*—fix a bug, drive a browser, operate a GUI. A complementary **social** ability, namely how well a model understands and forecasts the way real social events unfold, has barely been measured. We introduce **SocietyBench**, an end-to-end benchmark that takes a one-line event topic, collects Web news and social-media posts across five platforms, distills them into a date-indexed timeline that keeps *factual* events and a *public-opinion* layer separate, and then turns every cutoff date on that timeline into an audited bank of forecasting questions. Questions are scored on two orthogonal 100-point axes: **probability calibration** and **temporal accuracy**. Before any model sees a timeline, a three-phase procedure replaces every named entity and shifts every date by a per-event constant, turning a real arc into a *counterfactual social world*—structurally identical to what happened, but stripped of the surface labels a model could match against pre-training memory. On five heterogeneous events and 125 prediction points in Chinese and English editions, the strongest of six frontier LLMs reaches only 75.0 out of 100, against a trivial anchor of 50. The two axes come apart: a model can be calibration-strong but time-weak, or the reverse. Three agent frameworks built on a shared base model fail to improve on that base, and two model-free heuristics trail every LLM. Per-event gaps reach 21.4 points on a single axis, which is our main argument for evaluating on several events rather than one. All anonymized timelines, question banks, ground truth, and scoring code are released.

1. Introduction

The past two years produced a wave of LLM and agent benchmarks—SWE-bench, WebArena, OSWorld, τ-bench and their successors—that all ask a version of the same question: *can the model complete a task?* That axis has matured. It also covers only one side of what these systems are used for. The complementary ***social*** side—how well a model understands and forecasts the dynamics of a real social event—has hardly been quantified at all, and whether current models can do it remains mostly a matter of anecdote.

The benchmarks we have are not built to answer it. When a real, widely reported event is evaluated in its original form, a high score can reflect recognition of a memorized arc rather than genuine forecasting (Figure 1 contrasts the two regimes). Autocast (Zou et al., 2022) and ForecastBench (Karger et al., 2025) probe forecasting in a market-style, politico-economic setting, using a curated pool of expert-written "will X happen by Y?" questions. They are rigorous

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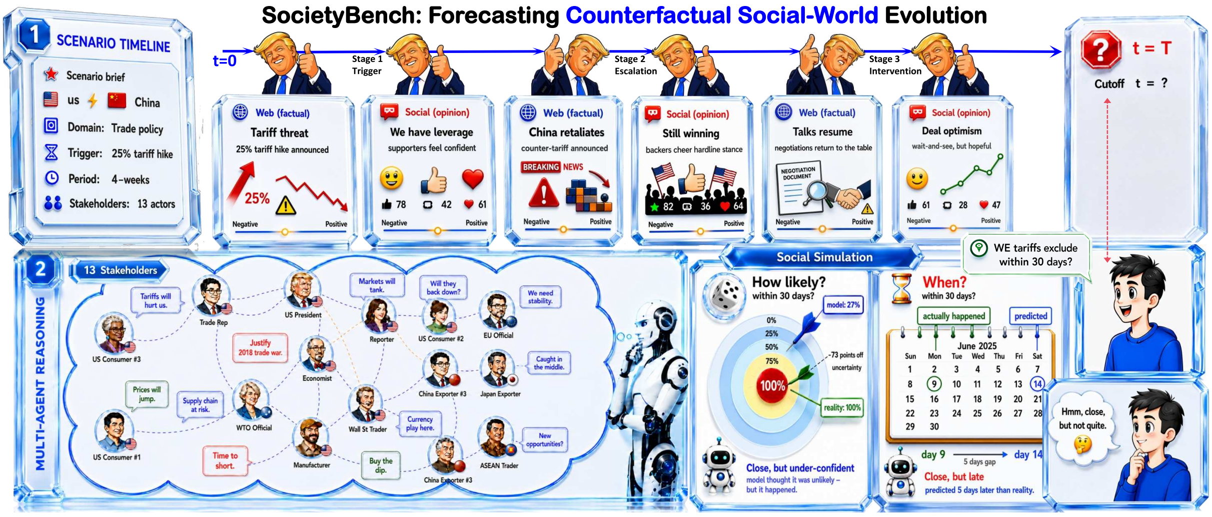

Figure 1 | **Why SocietyBench evaluates counterfactual forecasting rather than event recall.** Evaluated on a real, widely reported event in its original form, a model can score well simply by recognizing the event from pre-training data. SocietyBench replaces the entities and shifts the dates while preserving the causal and temporal structure, so the model has to forecast the future of an arc it cannot recognize.

probability benchmarks, but three gaps keep them out of the regime we care about. First, *the question types are narrow*—nearly every item is a binary yes/no call. Second, *contamination is avoided only by going live*: live questions have to wait on an unresolved future, which forfeits completed arcs, while historical questions already sit in pre-training corpora. Third, *public opinion is ignored*—yet in a social event, narratives flip and sentiment swings faster than the underlying facts change.

A second line of work is LLM-agent social simulation (Chuang et al., 2024; Man et al., 2025; Park et al., 2023; Piao et al., 2025; Yao et al., 2025a): populate a virtual town or forum with LLM agents playing personas, then watch the group behavior that emerges. These works ask whether an LLM can *perform* something that looks like a human society, not whether, given the opening of a real event, it can *predict* what happens next. The first question is about fidelity, the second about forecasting accuracy, and they are different tasks. Because a simulated environment has no real-world ground truth to score against, this line also cannot produce a quantitative leaderboard.

We argue that a benchmark for this regime has to satisfy four demands at once. It must (a) work on *real* events and real multimedia data, so the distribution matches how these systems are actually used; (b) integrate *both* factual reporting and public sentiment, since either alone misses half the dynamics; (c) build its ground truth without expert hand-annotation, so new events can be added as they unfold; and (d) score models with *calibrated* metrics that reward uncertainty-aware predictions rather than accuracy-style hits. No existing benchmark meets all four. In particular (b) and (c) are in tension, because social-media opinion is notoriously hard to align with objective ground truth. Our main design observation is that the conjunction becomes feasible if the benchmark is framed as a *retrospective* forecasting exam on an already-finished event, with the second half of the timeline playing the role of ground truth.

Concretely, we introduce **SocietyBench**, built on five heterogeneous social events spanning public controversy, geopolitics, technology policy, financial markets, and trade policy. For every event, an automated pipeline distills raw Web news and social-media posts into an anonymized, date-indexed timeline that separates *factual* events from a *public-opinion* layer, and then converts

Prefer uncertainty based predictions
rather than accuracy

every cutoff into a battery of audited forecasting questions (Section 3). The joint anonymization is more than a leakage filter. By replacing every named entity and shifting every date by a per-event offset, it turns each real arc into a *counterfactual social world*: structurally identical to what really happened, but without the surface labels a model could match against memory. A score therefore reflects *forward reasoning over social-world evolution* rather than recall of the specific real arc the timeline came from.

Our contributions:

- We release **SocietyBench**, a reproducible end-to-end benchmark for LLM and agent forecasting of real social events: five heterogeneous domains, dual-source timelines (Web news plus five social-media platforms), and a fully automated construction pipeline.
- We design a benchmarking protocol that pairs input-side **three-phase entity-and-date anonymization** with output-side **dual-axis scoring**—probability calibration and temporal accuracy—on calibrated 100-point metrics.
- We evaluate 6 frontier LLMs, 3 agent frameworks, and 2 non-LLM baselines end-to-end on 125 prediction points in two language editions. The strongest LLM reaches only 75.0 out of 100; the two axes dissociate; and per-event gaps reach 21.4 points on a single axis, which makes single-event evaluation unreliable.

2. Related Work

benchmarks that test whether the task was completed correctly or not

Task-completion agent benchmarks. Most existing LLM and agent benchmarks target the *task-completion* axis: SWE-bench (Jimenez et al., 2024) resolves real GitHub issues, WebArena (Zhou et al., 2024) runs long-horizon web tasks across four self-hosted apps, OSWorld (Xie et al., 2024) covers multi-application desktop workflows, τ-bench (Yao et al., 2025b) tests policy-bound dialogue agents, and GAIA (Mialon et al., 2024) poses general-assistant questions requiring tool use. All ask whether an agent can execute a prescribed sequence of operations to reach a *known* end state. SocietyBench is the social-dimension counterpart: instead of asking whether an agent can complete a task whose solution is fixed, we ask whether it can predict how a real-world social event will unfold.

goal of soceitybench vs
other benchmarks

benchmarks that test the forecast of events

Event and sentiment forecasting benchmarks. A separate line poses questions that can only be verified once events play out. Autocast (Zou et al., 2022), retrieval-augmented forecasting (Halawi et al., 2024), and ForecastBench (Karger et al., 2025) attach probability questions to dated news corpora. Context-Aware Sentiment Forecasting (Man et al., 2025) narrows to user-sentiment evolution during social-media events. MIRAI (Ye et al., 2024) forecasts country-level geopolitical events from a news database, and FutureX (Zeng et al., 2025) maintains a daily-refreshed live-web leaderboard. PolyBench (Cheng et al., 2026) is the only prior work that scores *time-of-event error in days*, but it does so on Polymarket short-horizon contracts. SocietyBench differs by targeting *medium-to-long-horizon social arcs* that carry both a *factual* and a *public-opinion* layer, rather than short-horizon binary contracts.

llms that simulate their own soceity

LLM-driven social simulation. A different line asks the inverse question: rather than *predicting* an event, can we *generate* a plausible society? Generative Agents (Park et al., 2023) first showed believable sandbox dynamics; AgentSociety (Piao et al., 2025) and OASIS (Yang et al., 2024) scaled the idea to tens of thousands and then a million agents; Opinion Dynamics (Chuang et al., 2024) and FDE-LLM (Yao et al., 2025a) study how opinions propagate through such networks. A

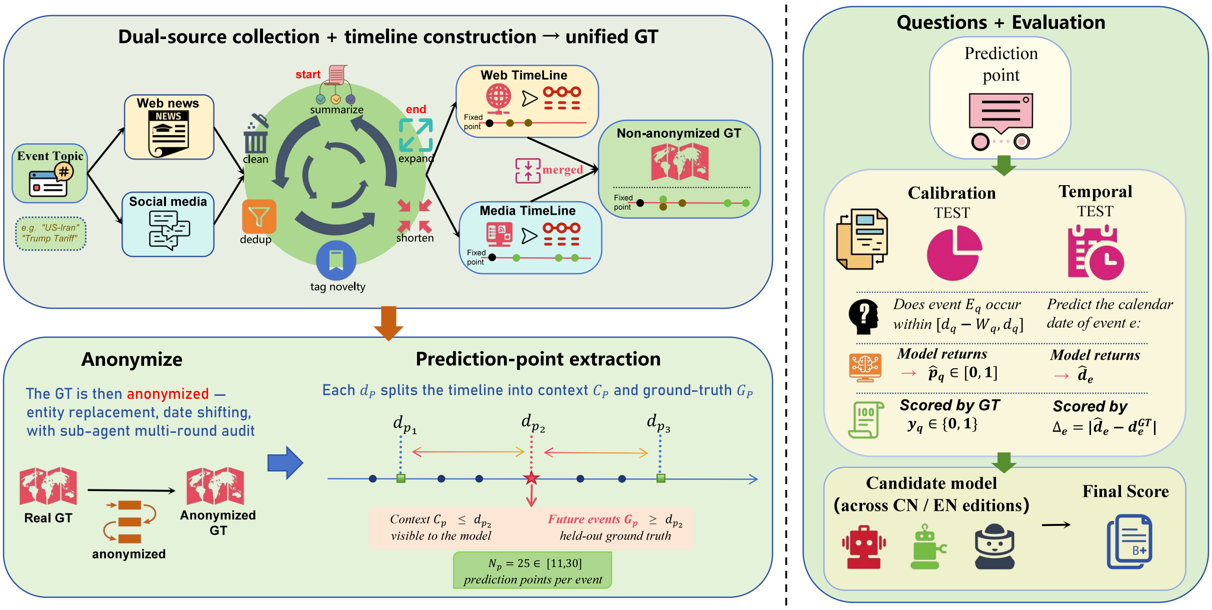

Figure 2 | SocietyBench methodology. Each retrospective cutoff d_P splits the anonymized timeline into a visible context C_P and a held-out set G_P. An audited bank emits a calibration set Q_P^{cal} and a temporal set Q_P^{time}, scored on two independent 100-point axes.

more recent cluster—SocioVerse (Zhang et al., 2025), MF-LLM (Mi et al., 2025), DualMind (Huang et al., 2026)—moves toward predictive validation, checking whether a simulated population's aggregate trajectory matches a real one. These target aggregate simulator fidelity rather than per-event forecast accuracy. The latter question—given a real unfolding social arc, can an LLM or agent system forecast it?—is what SocietyBench measures.

?

recent ones sheck simulator fidelity (?) rather than accuracy

3. Methodology

3.1. Task Formulation

The model plays a forecaster sitting at a cutoff date d_P. It sees everything that happened before d_P—the *context* C_P—and nothing after. It also cannot tell which real event it is looking at, because every named entity has been replaced and every date shifted (Section 3.2). It is then asked two kinds of questions about *what happens after* d_P (Figure 2). Together these form a prediction point

$$P = \left(d_P,\ C_P,\ G_P,\ Q_P^{\text{cal}},\ Q_P^{\text{time}}\right),$$

where the ground-truth set G_P is simply the timeline nodes from d_P onward.

whether an event occurs ?

- A **calibration question** $q = (E_q, d_q, W_q)$ asks whether event E_q occurs within the window $[d_q - W_q,\ d_q]$. The model returns a probability $\hat{p}_q \in [0,1]$ and is scored against the binary ground truth $y_q \in \{0,1\}$.

pred date of an event

- A **temporal question** asks for the calendar date $\hat{d}_e$ of a future event e in G_P. Error is measured in days.

Performance is summarized on two orthogonal axes—*probability calibration* and *temporal accuracy*—each converted to a 100-point score per event; full definitions are in Section 3.3. Every event is evaluated independently, and a bare LLM and a full agent system are run under exactly the same protocol.

Event	Nodes	Cal. questions	per point	Temporal events	in 0–90 d window
Wuhan Library	39	2,862	114.5	534	479
US-Iran	41	4,574	183.0	503	503
TikTok	127	6,697	267.9	571	485
SMCI	40	3,196	127.8	528	381
Trump Tariff	168	8,035	321.4	976	928
Total	**415**	**25,364**	**202.9**	**3,112**	**2,776**

#events 0-90days after cutoff date

Table 1 | **Per-event bank statistics**, per language edition; the two editions are one-to-one. *Nodes* is the timeline depth, and *in-window* counts the temporal events falling 0–90 d after their cutoff, which is the subset the temporal axis consumes. Every event contributes $N_P = 25$ prediction points.

3.2. Dataset and Anonymization

5 events from 5 domains

The benchmark covers five real events drawn from five domains—*public controversy*, *geopolitical conflict*, *technology policy*, *financial markets*, *and trade policy*—each spanning at least four weeks, so that the post-cutoff window admits a non-trivial horizon: a *university library harassment dispute* ("Wuhan Library"), a *U.S.–Iran conflict* ("US-Iran"), a *U.S. TikTok divestiture/ban ruling* ("TikTok"), a *Super Micro NASDAQ delisting crisis* ("SMCI"), and a *U.S. reciprocal-tariff escalation* ("Trump Tariff").

pipeline => takes event => creates {fact+opinion merged timeline, per event entity and date replacement table, set of prediction points, question bank for each point}

Each event is processed independently by the same pipeline, which produces a *fact + opinion* merged timeline, a per-event entity-and-date replacement table, an accepted set of prediction points, and a question bank for each prediction point. The pipeline runs automatically end to end; released artifacts additionally pass human acceptance audits, which gate quality but do not hand-author labels. After processing, timeline depth ranges from 39 to 168 nodes, with $N_P = 25$ prediction points per event. Per language edition, calibration questions range from 2,862 on *Wuhan Library* to 8,035 on *Trump Tariff*, and declared temporal events from 503 on *US-Iran* to 976 on *Trump Tariff*. Table 1 gives the per-event breakdown.

Anonymization. Frontier LLMs may have memorized a raw event—from its named entities down to the absolute calendar position of its arc—so each merged timeline is anonymized before any model sees it, obscuring *named entities* and *dates* jointly. Each event carries its own entity replacement table $\mathcal{R}$ and a single date offset $\delta \sim \mathcal{U}[-\delta_{\max}, +\delta_{\max}]$ applied uniformly to every date, which destroys absolute calendar position while leaving all inter-date gaps intact. We set $\delta_{\max} = 180\,\mathrm{d}$: a window that large crosses two seasons and a full quarterly reporting cycle, which defeats exact temporal lookup, while still keeping seasonal references coherent after the shift.

1. single date offset picked from a uniform dist

2. same offset applied to all events in one timeline

why memorisation of events by a model pays limited returns. (?)

Two checks bound what memorization can still buy. First, recalling the true calendar cannot pay by construction: re-scoring a model's own predictions as if it had answered on the *real* calendar collapses the temporal score to the same floor that an oracle knowing the true calendar attains against the shifted ground truth. Second, restoring real names and real dates outright moves the calibration axis by only +1.0 point on average, so binary outcome judgment does not lean on memorized identities. The three anonymization steps, summarized in Algorithm 1, are:

3 anonimization steps

- $\mathcal{A}$ **Substitution.** Replace entity strings via $\mathcal{R}$ using longest-match lookup, and shift every date by δ; a final scan repairs doubled substitutions.
- $\mathcal{B}$ **Reverse-identification audit.** A panel of three adversarial LLMs independently re-reads the anonymized text and tries to recover the original event and time period, reporting

3 llms audit and see whether they can recover the real event from anonymised text

Algorithm 1 Three-phase anonymization.

Require: Raw timeline $\mathcal{T}_{\text{raw}}$, replacement table $\mathcal{R}$, date offset δ, maximum audit rounds $K = 5$
Ensure: Anonymized timeline $\mathcal{T}_{\text{anon}}$, or event rejection

$\mathcal{T}^{(0)} \leftarrow$ replace entities in $\mathcal{T}_{\text{raw}}$ by longest-match lookup in $\mathcal{R}$
Shift every date in $\mathcal{T}^{(0)}$ by the same offset δ
Repair doubled substitutions and malformed date mentions
for $k = 1$ to K **do**
  Ask the adversarial auditor panel to identify the original event and time window
  Let $g_k \in \{\textsc{High}, \textsc{Mid}, \textsc{Low}\}$ be the strongest auditor confidence
  if $g_k = \textsc{Low}$ **then**
    break
  else if $g_k = \textsc{Mid}$ **then**
    Augment $\mathcal{R}$ and/or resample δ to remove residual clues
    Re-apply substitution and date shifting
  else
    return reject event
  end if
end for
Run paragraph-, event-, and narrative-level semantic-consistency checks
Repair any substitution damage while preserving anonymized entities and shifted dates
return $\mathcal{T}_{\text{anon}}$

confidence in $\{\textsc{High}, \textsc{Mid}, \textsc{Low}\}$. HIGH means the auditor named the specific event; MID means it named the event class; LOW means it was wrong. The augment-and-rerun loop follows Algorithm 1: an event is only released once no auditor still returns HIGH.

- C **Semantic-consistency pass.** Compare against the original at paragraph, event, and narrative level, repairing any damage introduced by substitution.

As argued in Section 1, the result is a counterfactual social world that a model cannot match against pre-training memory.

Bilingual release. Each event's question bank is released in Chinese and English in one-to-one correspondence. Both editions share the same entity placeholders and the same shifted dates, so neither edition can be used to de-anonymize the other.

3.3. Evaluation Protocol

Every model receives a **calibration score** S_{cal} and a **temporal score** S_{time}, both on a 0–100 higher-is-better scale.

every model gets 2 scores:
cal score + temp score

Calibration score. Each calibration question q carries a weight $w_q = w_q^{\text{time}} \cdot w_q^{\text{win}}$, the product of two factors:

- **Time factor** $w_q^{\text{time}} = 1/(1 + \alpha \cdot \Delta t_q)$ with $\alpha = 0.04$. The further a question resolves from the cutoff, in days Δt_q, the lower its weight, because far-future questions are noisier.
- **Window factor** $w_q^{\text{win}} = W_q/(W_q + 4)$. The shorter the question window W_q, the lower the weight, because very short windows admit only trivial predictions.

The weighted mean absolute error is then converted to a 100-point score:

$$\mathrm{wMAE}_{\mathrm{cal}} = \frac{\sum_q w_q \, |\hat{p}_q - y_q|}{\sum_q w_q}, \qquad S_{\mathrm{cal}} = 100 \cdot \max\big(0,\ 1 - \mathrm{wMAE}_{\mathrm{cal}}\big). \tag{1}$$

acc or score

1 - err = acc

A predictor that answers 50% to everything scores exactly 50, which anchors the scale. The choice of rule is not load-bearing: the scoring ablation of Section 4.3 re-scores every answer under proper scoring rules (Brier, log-loss) and the model ordering survives both.

the score here gets normalised later again.
And thus is less consequential

Temporal score. For every ground-truth event e the model predicts a date $\hat{d}_e$, with error $\Delta_e = |\hat{d}_e - d_e^{\mathrm{GT}}|$ in days. Each event falls in one of three 30-day buckets $[\ell_e, u_e]$—days 0–30, 31–60, or 61–90 after the cutoff—according to its offset t_e; day 0 is the cutoff date, which the context strictly precedes. A band of radius Δ_e centered on the true date, clipped to the bucket, gives the normalized miss:

1. Error calc from pred and true date. Δ_e
2. Band $\{B_e\}$ around true date of quant Δ_e
3. Gives normalised miss $\{m_e\}$

$$B_e = \min(u_e,\, t_e + \Delta_e) - \max(\ell_e,\, t_e - \Delta_e), \qquad m_e = \min\big(1,\ B_e/30\big). \tag{2}$$

Reusing the same time factor:

$$S_{\mathrm{time}} = 100 \cdot \max\left(0,\ 1 - \frac{\sum_e w_e^{\mathrm{time}} m_e}{\sum_e w_e^{\mathrm{time}}}\right). \tag{3}$$

A guesser that always answers the midpoint of its bucket scores roughly 50, so the two axes share a common trivial anchor.

Aggregation. Equation (1) and Equation (3) are computed independently on each event, giving a per-event pair $(S_{\mathrm{cal}}, S_{\mathrm{time}})$. A model's headline score is the *cross-event mean* of the five per-event scores, and per-event scorecards are reported alongside it.

headline score - cross event mean of 5 per event scores

4. Experiments

4.1. Implementation Details

5 api accessible LLMS + 3 custom agents

We evaluate six frontier, API-accessible LLMs—*GPT-5.5*, *Gemini-3.5-Flash*, *Claude-Opus-4.8*, *DeepSeek-V4-Pro*, *Kimi-K2.5*, and *Doubao-Seed-2.0-Pro*—together with three agent baselines built on the *Doubao* base model: LangGraph (LangChain Team, 2024) (plan-and-solve), AutoGen (Wu et al., 2024) (multi-agent debate), and MiroFish (Guo, 2026), a third-party open-source social-simulation forecaster built on OASIS (Yang et al., 2024). All systems are evaluated on the five events of Section 3.2, in both the Chinese and English editions.

on all 5 events of both eng and Chinese lang

All systems are queried deterministically (temperature 0) under a fixed reasoning budget, with the reasoning trace captured. No system is granted external web search beyond what its framework requires internally. Each system answers both question banks once at every prediction point. A question that still lacks a parseable answer after three retries is excluded from the primary score rather than silently defaulted, so no filled value reaches Table 2. The released materials cover the anonymized timelines, the question banks, and the ground truth; per-question model responses are not part of the release.

Temperature controls the randomness of text that is generated by LLMs during inference.

	• Wuhan Lib.		• US-Iran		• TikTok		• SMCI		• Trump Tariff		Avg.		
System	Cal.	Time	Cal.	Time	Cal.	Time	Cal.	Time	Cal.	Time	Cal.	Time	Overall
GPT-5.5	74.7	55.6	66.3	63.2	78.7	84.4	74.9	84.2	81.7	85.4	**75.3**	**74.6**	**75.0**
Gemini-3.5-Flash	71.0	59.6	68.9	58.5	79.8	79.4	72.3	73.3	77.8	66.1	**73.9**	**67.3**	**70.6**
Claude-Opus-4.8	59.0	63.2	55.0	62.3	64.2	72.6	60.8	70.4	73.6	81.7	**62.5**	**70.0**	**66.3**
DeepSeek-V4-Pro	63.4	58.6	55.9	64.9	65.1	66.7	60.8	65.2	67.5	69.0	**62.5**	**64.9**	**63.7**
Kimi-K2.5	64.4	59.4	58.4	65.6	62.5	63.9	64.8	64.9	66.4	64.0	**63.3**	**63.6**	**63.4**
Doubao-Seed-2.0-Pro	65.5	58.3	57.9	60.3	67.0	64.2	60.5	65.0	70.3	65.0	**64.3**	**62.6**	**63.4**

Table 2 | Main leaderboard. Per-event and cross-event calibration / temporal means for six frontier LLMs; each cell averages the Chinese and English editions. A uniform 50% predictor scores exactly 50 on the calibration axis, and a bucket-midpoint guesser scores about 50 on the temporal axis.

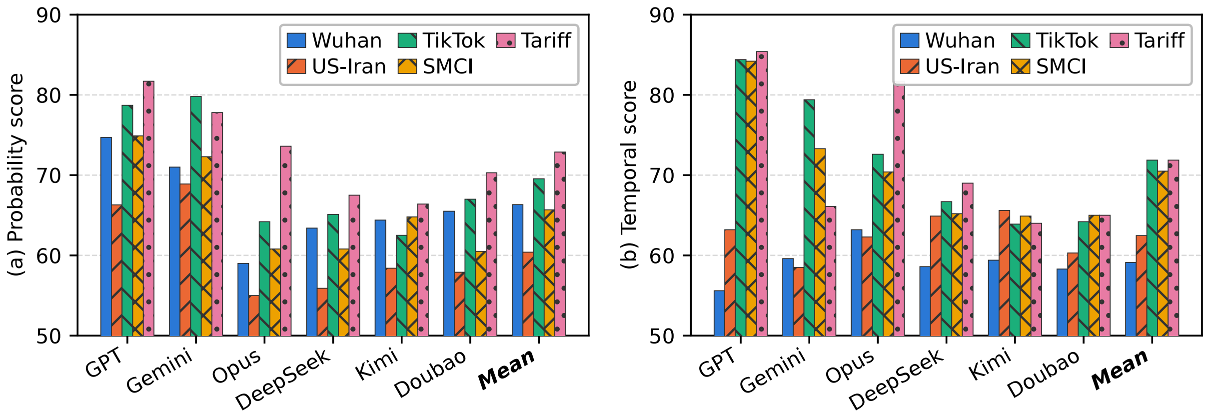

Figure 3 | Per-event scores. *Mean* is the six-LLM average on each axis.

4.2. Main Results

Table 2 reports the main leaderboard, with per-event scorecards on the left (visualized in Figure 3) and the cross-event mean on the right; every cell averages the two language editions.

Two findings stand out.

The spread is wide at the top and tight at the bottom. *GPT-5.5* leads at 75.0, followed by *Gemini-3.5-Flash* at 70.6 and *Claude-Opus-4.8* at 66.3, while the bottom three sit within 63.4–63.7 of one another—an 11.3-point lead (+17.7%) over that cluster. The lead is not an artifact of one lucky event: leaving any single event out and re-averaging still leaves GPT-5.5 ahead by at least 2.4 points, and it wins outright on four of the five events.

The two axes come apart. *GPT-5.5* leads both (75.3 calibration / 74.6 temporal), but *Gemini-3.5-Flash* is calibration-strong and time-weak (73.9 vs. 67.3) while *Claude-Opus-4.8* is the mirror image (62.5 vs. 70.0). Neither axis is a proxy for the other, which is the practical reason we report both rather than a single blended number. Even the leader recovers only about half the headroom above the trivial anchor of 50, so the benchmark is far from saturated.

Stress-case events. Two events stand out as stress cases (Table 3; Figure 4 traces both qualitatively). On *Trump Tariff*, *Kimi-K2.5* scores lowest on both axes (66.4 / 64.0) and the temporal gap to *GPT-5.5* reaches 21.4 points (+33%)—the largest single-event, single-axis gap in the whole table—on the very event where most models post their best calibration scores. On *SMCI* the

	Event statistics				Mean		GPT-5.5		Kimi-K2.5		Gap Δ	
Event	N_P	Arc	#Cal.	#Time	Cal.	Time	Cal.	Time	Cal.	Time	Cal.	Time
SMCI	25	468d	3,196	528	65.7	70.5	74.9	84.2	64.8	64.9	$\mathbf{10.1}_{16\%}$	$19.3_{30\%}$
TRUMP-TARIFF	25	2,930d	8,035	976	72.9	71.9	81.7	85.4	66.4	64.0	$15.3_{23\%}$	$\mathbf{21.4}_{33\%}$

Table 3 | Stress-case comparison. *GPT-5.5* (strongest overall) versus *Kimi-K2.5* (weakest on the temporal axis); *Mean* is the six-LLM mean; *Gap* Δ is GPT minus Kimi, with subscripts giving the gap as a percentage of Kimi's score. *Arc* is the span covered by the event's prediction points, and #Time counts declared events (only those 0–90 d after a cutoff are scored).

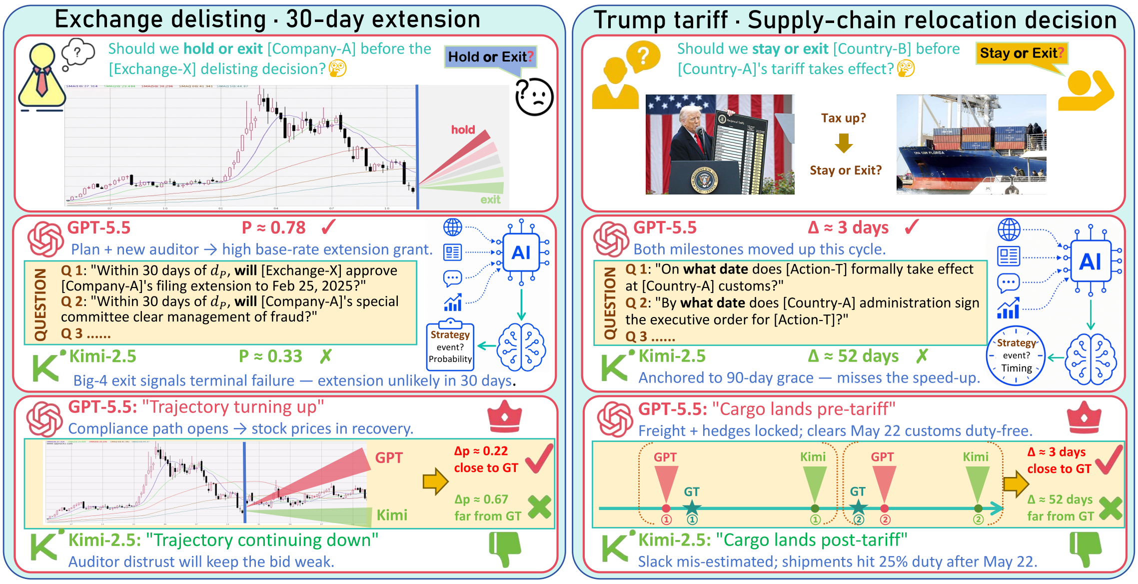

Figure 4 | Qualitative model comparison. The SMCI extension decision (left) and the tariff decision (right), *GPT-5.5* (✓) versus *Kimi-K2.5* (✗). GPT commits to a calibrated probability and lands close to the true date; Kimi under-commits and anchors far away.

same GPT–Kimi gap is double-digit on both axes (+10.1 / +19.3), which is consistent with an event built around company-specific numerical disclosures that reward precise uncertainty handling. The practical implication is the one we flag in Section 1: a benchmark of this kind reports a much noisier picture if it is run on a single event—two of our five events would each, on their own, tell a materially different story about how far apart the models are.

4.3. Ablation Study → remove some stuff, and see the impact on value

We ablate four design choices: question-bank composition, the calibration scoring formula, true/false composition, and cutoff position.

Question-bank composition. Table 4 drops one subset of the bank at a time and re-scores. The four audited types are: **A**, does the same event occur within windows growing from 7 to 90 days; **B**, one fact re-asked at rising levels of detail; **C**, true outcomes paired with counterfactual variants; **D**, events built to be false. Dropping A inflates the six-LLM mean the most (+7.2), so A carries most of the benchmark's difficulty. It targets a specific failure: models over-bet on a currently hot narrative continuing in the near term, which is exactly what a growing-window question probes. Dropping D moves the mean the other way (−3.4), making it the easiest subset—models are good at rejecting invented events. B and C shift the mean by at most 0.8, so

A. after dropping A, there are no events that repeat in the far future. And models score higher. The only events left are that repeat in the near future (within 7 days of cut-off).
{models say that current hot topic would surely continue in the near future and thus score higher as only near repeating events are left.

Ablation analysis on ques bank composition

D. Events that are built to be false. If you remove them the only events left are ones that could surely be true. Now the mean decreases, meaning difficulty increased.
{Therefore models were able to recognise made to false events very easily}

Question-bank composition	Doubao	GPT	Mean
– A-type (time-gradient, 47%)	$70.7_{+6.4}$	$83.4_{+8.1}$	$\mathbf{74.2}_{+7.2}$
– B-type (specificity, 23%)	$62.9_{-1.4}$	$74.1_{-1.2}$	$\mathbf{66.2}_{-0.8}$
– C-type (variants, 12%)	$63.9_{-0.4}$	$74.8_{-0.5}$	$\mathbf{66.6}_{-0.4}$
– D-type (designed-false, 18%)	$62.1_{-2.2}$	$72.2_{-3.1}$	$\mathbf{63.6}_{-3.4}$
Full $\text{bank}_{\text{Cal.}}$ (ours, ~25k Q/ed.)	**64.3**	**75.3**	**67.0**
– Near events (0–30 d, 52%)	$62.0_{-0.5}$	$78.0_{+3.4}$	$\mathbf{67.7}_{+0.5}$
– Mid events (31–60 d, 29%)	$62.8_{+0.3}$	$74.4_{-0.2}$	$\mathbf{67.4}_{+0.2}$
– Far events (61–90 d, 18%)	$62.5_{-0.1}$	$73.5_{-1.1}$	$\mathbf{66.8}_{-0.4}$
Full $\text{bank}_{\text{Time}}$ (ours, ~3.1k ev./ed.)	**62.6**	**74.6**	**67.2**

Table 4 | Question-bank composition. The upper block drops a question type, the lower block a horizon bucket (shares rounded); subscripts are relative to the full bank.

Calibration scoring formula	Doubao	GPT	Gap
Equal weights (w=1)	$64.2_{-0.1}$	$74.2_{-1.1}$	10.0
Linear time-decay (1−d/365)	$65.4_{+1.1}$	$76.0_{+0.7}$	10.6
No time-decay (α=0)	$64.8_{+0.5}$	$75.0_{-0.3}$	10.2
Steeper decay (α=0.08)	$63.8_{-0.5}$	$75.1_{-0.2}$	11.3
Steeper decay (α=0.16)	$63.1_{-1.2}$	$75.0_{-0.3}$	11.9
Brier score (rescaled)	$57.4_{-6.9}$	$70.8_{-4.5}$	13.4
Log-loss (rescaled)	$45.4_{-18.9}$	$65.0_{-10.3}$	19.6
Hyperbolic score mapping	$73.4_{+9.1}$	$80.1_{+4.8}$	6.7
Weighted MAE (ours, α=0.04)	**64.3**	**75.3**	**11.0**

Table 5 | Calibration scoring formula. Alternative weights, metrics, and mappings; *Gap* is GPT minus Doubao.

True/false Q's	Doubao	DSeek	Opus	Gemini	GPT	Mean (6)
True	59.8	49.3	46.4	72.0	73.5	**60.3**
False	71.6	79.9	84.3	76.6	78.3	**76.1**
Bias	**+11.8**	**+30.6**	**+38.0**	**+4.7**	**+4.8**	**+15.9**

Table 6 | True/false-question bias (calibration axis). Models deny false events better than they confirm true ones; *Bias* is false minus true.

Cutoff position	Doubao	Opus	Gemini	GPT	Mean (6)
Early	63.4	62.7	75.1	75.5	**66.7**
Middle	$66.1_{+2.7}$	$63.5_{+0.8}$	$73.6_{-1.5}$	$75.4_{-0.1}$	$\mathbf{67.6}_{+0.9}$
Late	$66.6_{+0.5}$	$68.4_{+4.9}$	$69.6_{-4.0}$	$77.4_{+2.0}$	$\mathbf{68.7}_{+1.1}$
Gradient	**+3.2**	**+5.7**	**−5.5**	**+1.9**	**+2.0**

Table 7 | Cutoff gradient (calibration axis). Later prediction points see more history and score higher; *Gradient* is late minus early.

neither changes the ranking on its own. On the temporal side, dropping any horizon bucket moves the mean by at most 0.5, so that axis is robust to the horizon mix.

Calibration scoring formula. Table 5 re-scores the same answers under eight alternative calibration rules. Removing the time decay, steepening it, or dropping the question weights altogether moves scores by at most 1.2 points, so the score is insensitive to the exact weighting. Replacing MAE with Brier or log-loss deflates Doubao by 6.9 and 18.9 points respectively; a hyperbolic mapping inflates both models and shrinks their gap to 6.7. Under every rule the GPT–Doubao gap stays positive, so the ranking does not depend on the choice of rule. And against the suspicion that a designer picks the metric that flatters the result: the proper scoring rules (Brier, log-loss) *widen* the gap rather than narrowing it, because they punish confident errors harder and weaker models make more of them. Our weighted MAE is the gentler choice. We keep it because it sits in the middle of the score range while up-weighting the near-cutoff questions, where forecasting is hardest, and down-weighting trivially short windows.

1. proper scores Brier, log loss. Give wider gap of error b/w models
2. MAE is a gentler choice

MAE advantages

True/false-question bias. Table 6 splits the calibration bank by answer: events that did occur, versus the false variants of types A–C and the designed-false D set. Models are much better at *denying false events* than at *confirming true ones*—six-LLM means of 76.1 against 60.3, and a mean per-model bias of +15.9. This is a disposition, not raw strength. The two leaders are near balanced (*GPT-5.5* +4.8, *Gemini-3.5-Flash* +4.7); *Claude-Opus-4.8* is the extreme at +38.0, very good at rejecting what did not happen and poor at endorsing what did. A different true/false ratio would reward that disposition, so we keep the bank near 1:1—the axis should measure forecasting, not default skepticism.

models are much much better at rejecting false events than at confirming true ones.

	• Wuhan Lib.		• US-Iran		• TikTok		• SMCI		• Trump Tariff		Avg.		
System	Cal.	Time	Cal.	Time	Cal.	Time	Cal.	Time	Cal.	Time	Cal.	Time	Overall
Doubao (base)	65.5	58.3	57.9	60.3	67.0	64.2	60.5	65.0	70.3	65.0	**64.3**	**62.6**	**63.4**
LangGraph	62.7	55.8	55.9	60.4	65.8	64.7	56.6	65.5	69.0	64.9	**62.0**$_{-2.3}$	**62.3**$_{-0.3}$	**62.1**$_{-1.3}$
AutoGen	59.2	57.2	52.6	61.4	63.9	64.7	50.3	64.5	66.6	65.0	**58.5**$_{-5.8}$	**62.5**$_{-0.1}$	**60.5**$_{-2.9}$
MiroFish	66.2	57.5	57.1	59.2	65.5	65.5	59.2	65.7	70.3	64.4	**63.7**$_{-0.6}$	**62.4**$_{-0.2}$	**63.1**$_{-0.3}$

Table 8 | **Agent orchestration.** Three agent frameworks running on the *Doubao* base model (first row); subscripts give the change relative to the base.

Cutoff gradient. Table 7 splits each event's 25 prediction points into early, middle, and late terciles. Later cutoffs, which see more of the arc, score higher: 66.7 → 67.6 → 68.7. This is the "closer is easier" property a forecasting benchmark should show, and a basic sanity check—flat or inverted scores would mean the questions are not measuring forecasting from context. The trend holds for five of the six LLMs, reaching +5.7 on *Claude-Opus-4.8*. *Gemini-3.5-Flash* is the exception at −5.5, best at the earliest cutoffs: a quirk of that model, not of the benchmark, and worth knowing before reading any single-cutoff evaluation of it.

scores are better when llms are fed more data. i.e cutoff is much higher

4.4. Agents and Non-LLM Baselines

Two questions put the leaderboard in context: does agent orchestration lift a base LLM, and how much of the LLMs' performance can a model-free heuristic recover?

Agent orchestration. Table 8 runs three orchestration frameworks on the *Doubao* base. None improves on the base: LangGraph loses 1.3 points overall, AutoGen 2.9, and MiroFish 0.3. The losses concentrate on the calibration axis, reaching −5.8 for AutoGen, while temporal scores stay within 0.3 of the base—orchestration perturbs probabilities far more than it perturbs dates. Read against Table 2, the base model, not the scaffold, determines where a system lands; no agent setup approaches the leading single LLMs. This is a negative result worth stating plainly, because the intuition that a debate or plan-and-solve wrapper should help on a reasoning-heavy forecasting task is a natural one, and it does not hold here.

Non-LLM baselines. Two model-free heuristics anchor the scale. A *frequency baseline* answers each calibration question with the empirical base rate of similar events in the visible context, and a *momentum baseline* does the same over a 7-day rolling window, implementing exactly the near-term-continuation assumption that the ablation above identifies as the dominant LLM failure mode. They score 53.8 and 54.2 on the calibration axis—8 to 21 points below every LLM in Table 2, and only about 4 points above the trivial anchor. Neither emits dates, so their temporal score is recorded at the bucket-midpoint anchor of 50. Two things follow: the leaderboard reflects predictive signal rather than an artifact of the scoring formula, and the failure mode the LLMs fall into carries almost no predictive value as a deliberate strategy.

non llm baseline scores are created using 2 models:
1. freq baseline : Ans based on freq/rate of same events within visible context
2. momentum baseline : same over a short 7 day rolling window. That means freq events within near term are more likely to occur.

5. Conclusion

We presented **SocietyBench**, an end-to-end benchmark that anonymizes real social events into *counterfactual social worlds*—arcs that can be predicted only by genuine social-dynamic reasoning, not by pre-training recall—and scores models on probability calibration and temporal gpt 5.5

freq & mom baselines

accuracy. Across five heterogeneous events and 125 prediction points in two language editions, the strongest of six LLMs reaches only 75.0 against a trivial anchor of 50; three agent frameworks fail to beat their base LLM; and two non-LLM baselines trail every LLM. Per-event gaps reach 21.4 points on a single axis, which is our main argument for multi-event over single-event evaluation. biased results if using only a single event

Useful next steps are stronger anonymization to separate forecasting from memory even more cleanly, a broader event pool, finer discrimination among near-peer models, and community-contributed events and systems.

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